

## SECTION I (50 marks)

*Answer all the questions in this section in the spaces provided.*

- 1 Without using a mathematical table or calculator, evaluate:

$$\sqrt{\frac{11}{12}} - \frac{1}{3} \div 1\frac{1}{2}$$

- 2 Baraka earns Ksh 210 per hour working at a supermarket. The employer changed earned per hour in the ratio of 8:7.

Determine the amount that Baraka would earn in  $10\frac{1}{2}$  hours at the new rate.

- 3 Solve for  $x$  in the equation.

$$4^{3x} \times 8 = \left(\frac{1}{32}\right)^{2x-3}$$



A clock which loses 18 seconds every hour was set to read the correct time at 8.00 am on Monday. Determine the time, in 12 hour system, the clock will read on the following Saturday at 11.20 am. (3)

|         |        |         |  |
|---------|--------|---------|--|
| VS - 12 | ST - 1 | 11 - 01 |  |
|         |        |         |  |
| 21      | 2      | 11      |  |

In the following figure, Line  $AB = 5$  cm is a side of a triangle  $ABC$  in which  $\angle BAC = 60^\circ$  and line  $BC = 10$  cm. Using a pair of compasses and a ruler, complete triangle  $ABC$  and measure the size of angle  $ABC$ . (3)



A shopkeeper bought an item from a wholesaler. If the shopkeeper sells the item for Ksh 2 340, he would make a profit of Ksh  $3x$ . If the shopkeeper sells that item for Ksh 2 340, he would make a loss of Ksh  $2x$ . Determine the amount that the shopkeeper paid for the item.

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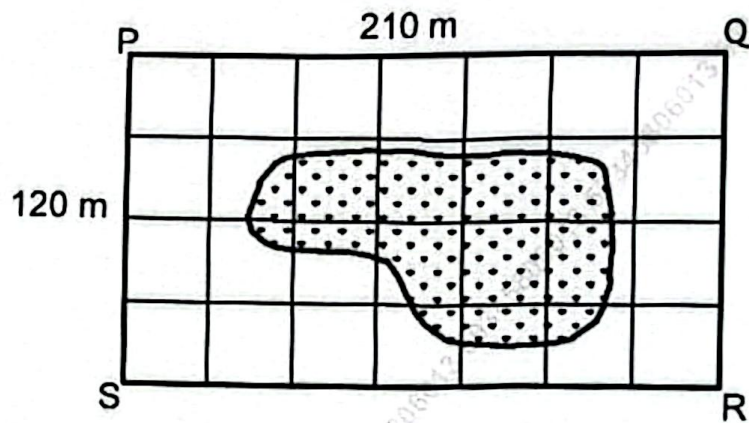
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4 Solve  $-1 \leq \frac{5-2x}{3} < 2x-1$ , giving the answer as a combined inequality.

- 5 The following figure represents a rectangular farm, PQRS, of length 210 m and drawn on a grid of 1 cm squares. The dotted area inside the farm represents a flood

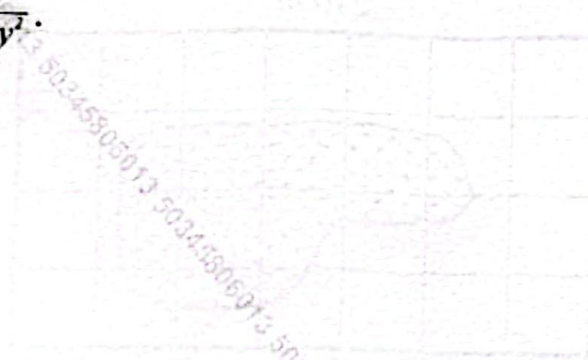


Estimate in  $\text{m}^2$ , the area of the farm that is not flooded.

A relief organization donated 240 kg of maize and 150 kg of beans to needy families. Each family received exactly the same quantity by mass of either maize or beans. No family received both. Determine the least possible number of the needy families. (3 marks)

Simplify  $\frac{x^2 - 4y^2}{x^2 + 4xy + 4y^2}$ .

(3 marks)



The area of a sector of a circle is  $550 \text{ cm}^2$ . The sector is curved to form an open cone of radius 7 cm. Calculate the height of the cone. (4 marks)

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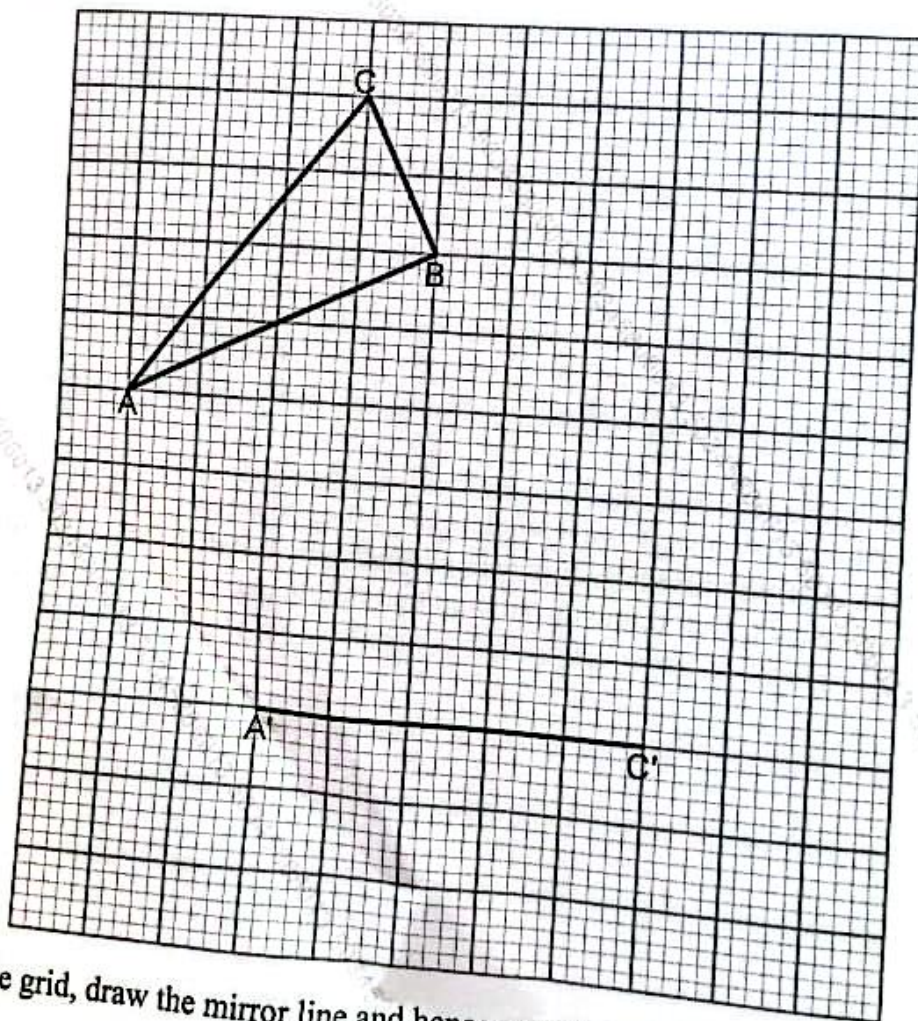
- 12 The following table shows the frequency distribution of marks scored by students in a Mathematics test. The frequencies for two classes are not shown. The table also shows height of each rectangular bar of a histogram drawn to represent the information on the scored by students.

| Marks         | 10 - 14 | 15 - 19 | 20 - 29 |
|---------------|---------|---------|---------|
| Frequency     | 7       |         |         |
| Height of bar | 1.4     | 2       | 1.5     |

Calculate the missing frequencies.

(3)

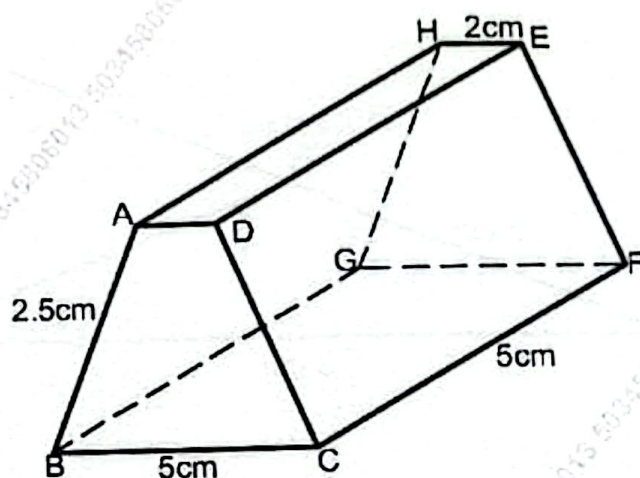
- 13 On the following grid, line  $A'C'$  is part of  $\triangle A'B'C'$ . Triangle  $A'B'C'$  is the image of  $\triangle ABC$  after a reflection along a mirror line.



On the same grid, draw the mirror line and hence complete  $\triangle A'B'C'$ .

(3 mar

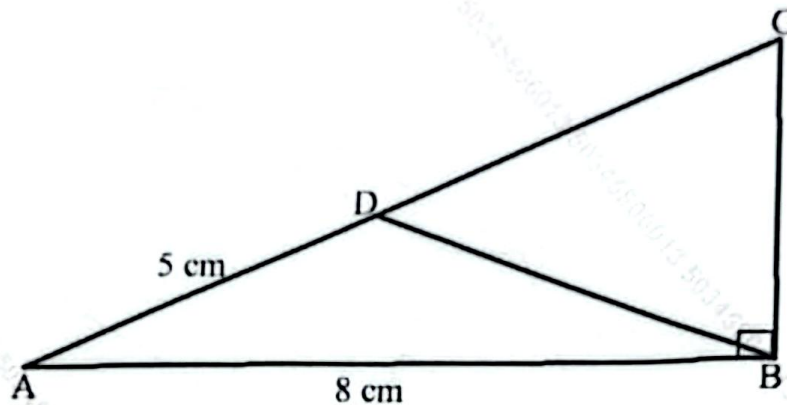
- 14 The figure below represents a prism ABCDEFGH. The cross section of the prism is a trapezium.  $BC = CF = 5$  cm,  $AD = HE = 2$  cm and  $AB = DC = 2.5$  cm.



Draw a labelled net of the prism.



- 15 In the following figure, triangle ABC is right angled at B and  $AB = 8$  cm. D is a point on AC such that  $AD = 5$  cm and the area of triangle ABD =  $10 \text{ cm}^2$ .



Calculate the length of side BC.

(4 marks)

- 16 The quadratic curve  $y = 3x^2 - 4x$  passes through the point  $P(1, -1)$ . Determine the equation of a tangent to the curve at point P leaving the answer in the form  $ax + by = c$ , where  $a$ ,  $b$  and  $c$  are scalars.

(3 marks)

**SECTION II (50 marks)**

*Answer only five questions in this section in the spaces provided.*

- 17 The road connecting town A to town B is 160 km long. A lorry, travelling at an average speed of 45 km/h, left town A for town B at 11.50 am. At that same time, a car travelling at an average speed of 75 km/h, left town B for town A. The two vehicles met at C, a town along the road.

(a) Determine:

(i) the time when the two vehicles met. (4 marks)

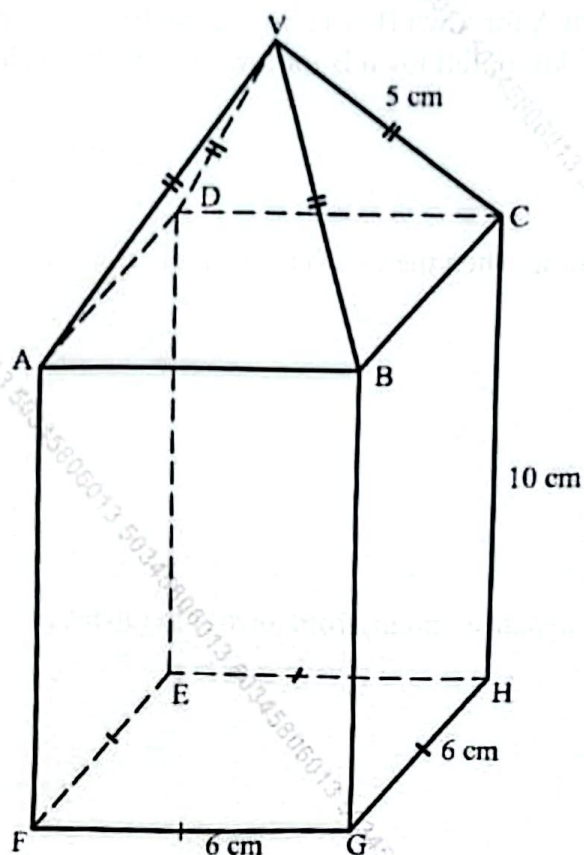
(ii) the distance, in km, from town A to town C. (2 marks)

- (b) The car stopped at town C for a period of 1 hour 40 minutes. The lorry continued with its journey at the same speed of 45 km/h. After the stop, the car left town C for town A and arrived at its destination at the same time as the lorry arrived at B. Determine the average speed of the car for the journey from town C to town A.

(4 marks)



- 18 Figure VABCDEFGH shows a solid consisting of a right pyramid mounted on a rectangular block. The right pyramid and the rectangular block have an identical square base of length 6 cm. The height of the rectangular block is 10 cm while  $VA = VB = VC = VD = 5$  cm.



- (a) Calculate the volume of the solid.

(4 marks)



- (b) The solid is cut into two identical halves along the plane of symmetry VAFHCV. Calculate the surface area of one of the pieces. (6 marks)

19 Two lines,  $L_1$  and  $L_2$  are parallel. Line  $L_3$  is perpendicular to both lines  $L_1$  and  $L_2$ .

(a) (i) Line  $L_1$  passes through points  $A(-3, 6)$  and  $B(3, 9)$ . Find the gradient of line  $L_1$ . (2 marks)

(ii) Line  $L_2$  intersects the  $x$ -axis at point  $C(-5, 0)$ . Determine the equation of  $L_2$  in the form  $y = mx + c$ . (2 marks)

(b) Line  $L_3$  intersects lines  $L_1$  and  $L_2$  at points  $A(-3, 6)$  and  $P$  respectively.

(i) Determine the equation of  $L_3$  in the form  $y = mx + c$ . (3 marks)

(ii) Determine the coordinates of point  $P$ . (3 marks)

- 21 In the grid provided, the coordinates of points P and Q are (3, 3) and (8, 5) respectively. Vector  $\overrightarrow{PQ}$  is also shown.



- (a) Express  $\overrightarrow{PQ}$  as a column vector.

(2 marks)

- (b) Two vectors,  $\mathbf{a} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$  and  $\mathbf{b} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$  are such that  $k\mathbf{a} + h\mathbf{b} = \overrightarrow{PQ}$  where  $k$  and  $h$  are scalars.

- (i) Determine the values of  $k$  and  $h$ .

(4 marks)

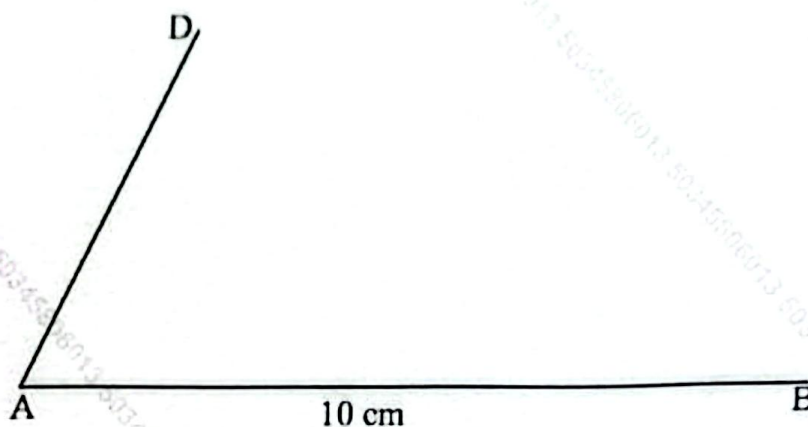


15

- (ii) On the same grid, from point P, represent by accurate drawing, the sum of vectors  $ka$  and  $hb$ . (2 mark)

- (c) Determine the magnitude of vector  $ka$ . (2 m

- 22 In the following figure, lines  $AB = 10$  cm and  $AD = 5$  cm are part of a trapezium  $ABCD$  in which line  $AB$  is parallel to line  $DC$ . Angle  $DAB = 65^\circ$ .



- (a) On the same figure, using a ruler and a pair of compasses only:
- complete trapezium  $ABCD$  given that  $\angle ABD = 67.5^\circ$ . Measure the length of line  $DC$ ; (4 marks)
  - drop a perpendicular line from  $D$  to meet  $AB$  at  $E$ . Measure the length of line  $DE$ ; (2 marks)
  - draw an arc of a circle, centre  $A$  and radius  $5$  cm. The clockwise directed arc should join  $D$  to a point  $F$  on line  $AB$ . (1 mark)
- (b) Calculate the area of trapezium  $ABCD$  that lies outside the arc drawn in (a)(iii). (Take  $\pi = 3.142$ ) (3 marks)

The quantity of petrol in litres used by 40 boda boda riders on a particular day is as follows.

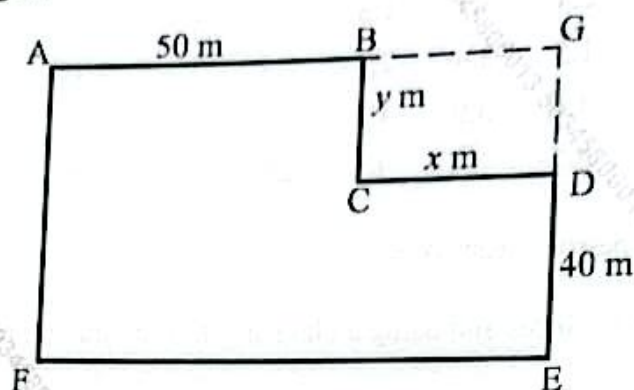
|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| 1.7 | 2.9 | 2.1 | 2.8 | 3.6 | 2.1 | 1.6 | 2.8 |
| 4.0 | 2.4 | 3.1 | 1.8 | 2.5 | 1.9 | 2.6 | 3.2 |
| 3.1 | 1.5 | 3.4 | 2.9 | 2.9 | 3.4 | 2.1 | 2.6 |
| 3.4 | 4.1 | 3.9 | 4.3 | 3.7 | 2.8 | 2.3 | 2.7 |
| 3.3 | 2.7 | 2.4 | 3.5 | 2.0 | 1.5 | 1.8 | 2.3 |

The cost of 1 litre of petrol is Ksh 160.

- (a) Starting with 1.5 litres and using a class size of 0.5, draw a frequency distribution table for the data. (2 marks)
- (b) Use the frequency distribution table in (a) to estimate:
- (i) the mean amount of money spent on petrol by the boda boda riders on that day. (4 marks)
- (ii) the median amount of money spent on fuel by the median boda boda rider. (4 marks)



- 24 Figure ABCDEF represents a plot of land. Length  $AB = 50$  m,  $BC = y$  m,  $CD = x$  m and  $DE = 40$  m. The figure also shows an adjacent plot of land BCDG. All the angles in figure are right angles.



The owner of plot ABCDEF later purchased plot BCDG. Given that the perimeter of plot BCDG is 60 m:

- (a) (i) Write an expression for  $y$  in terms of  $x$ . (2 m)
- (ii) Write down a simplified expression in  $x$  for the area of the entire plot of land AGEF. (3 m)
- (b) (i) Determine the dimensions of plot BGDC that would maximize the size of the entire plot of land AGEF. (3 m)
- (ii) Determine, in  $m^2$ , the maximum possible area of plot AGEF. (2 m)

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